Project Four: Fund X Factor Model Analysis

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July 2026

Data and Method

The analysis uses Fund X's net-of-fees monthly returns from January 2017 through December 2024 and the official monthly q5 factor file from `https://global-q.org/factors.html`. The 96 observations are merged by year and month. Every factor and return is measured in percent per month. The dependent variable is Fund X's excess return, $r_{X,t} - r_{f,t}$, where $r_{f,t}$ is the `R_F` observation supplied in the q5 file.

Three time-series regressions are estimated: the CAPM with R_{MKT}; the Hou–Xue–Zhang q-factor model with R_{MKT}, R_{ME}, R_{IA}, and R_{ROE}; and the augmented q5 model, which adds R_{EG}. Coefficients are ordinary least squares estimates. The reported t-statistics use Newey–West heteroskedasticity- and autocorrelation-consistent covariance estimates with the assignment-mandated Bartlett lag length $L = 4$.

Table 1: Full Regression Results

Table 1: Fund X Factor Regressions: CAPM, q-Factor, and q5

	Model 1 CAPM	Model 2 q-factor	Model 3 q5
α (monthly, %)	3.015***	3.041***	2.965***
	[4.93]	[5.24]	[5.48]
β_{MKT}	-0.056	-0.022	0.010
	[-0.50]	[-0.26]	[0.14]
β_{ME} (size)	–	-0.222	-0.093
	–	[-1.29]	[-0.69]
β_{IA} (investment)	–	0.122	0.273
	–	[1.12]	[1.39]
β_{ROE} (profitability)	–	-0.161	-0.350
	–	[-1.26]	[-1.88]
β_{EG} (expected growth)	–	–	0.366
	–	–	[1.07]
R^2	0.006	0.040	0.069
Adjusted R^2	-0.005	-0.002	0.017
N	96	96	96

Note: Sample: January 2017 – December 2024 (N = 96 monthly observations). Dependent variable: Fund X monthly excess return ($r_X - r_f$). q5 factor returns (R_{MKT}, R_{ME}, R_{IA}, R_{ROE}, R_{EG}) are from Hou, Mo, Xue and Zhang (2021) via global-q.org. Standard errors are Newey–West HAC with $L = 4$ lags (Newey and West, 1987); t-statistics in brackets. No coefficient is asterisked unless its $|t| \geq 1.96$. *** denotes $p < 0.001$.

Table 2: Alpha Summary

Table 2: Summary of Alpha Estimates Across Factor Models

Model	Monthly α	NW t-stat	Annual α	R^2
CAPM (1-factor)	3.015***	4.93	42.82%	0.006
q-factor (4-factor)	3.041***	5.24	43.27%	0.040
q5 (5-factor)	2.965***	5.48	42.00%	0.069

Note: *** $p < 0.001$. Annual alpha computed as $(1 + \text{monthly\_alpha}/100)^{12} - 1$. Sample: January 2017 – December 2024, $N = 96$. NW t-statistics use $L = 4$ lags.

Results

Across all three specifications, Fund X retains a large positive intercept. The CAPM monthly alpha is 3.01% with a Newey–West t-statistic of 4.93. The q-factor estimate is 3.04% ($t = 5.24$), and the q5 estimate is 2.97% ($t = 5.48$). The full range across specifications is only 7.6 basis points, while the change from CAPM to q5 is -5.0 basis points. The corresponding compounded annual alpha estimates range from 42.00% to 43.27%. All three intercepts have $p < 0.001$. The t-statistic rises as factors are added, despite the slight decline in the q5 point estimate, because the HAC standard error becomes smaller. The additional factors therefore do not absorb Fund X's average excess return.

The market loading is economically small and statistically insignificant in every model: it moves from -0.056 in the CAPM to 0.010 in q5. In the q5 model, the size loading is -0.093 ($t = -0.69$), the investment loading is 0.273 ($t = 1.39$), and the expected-growth loading is 0.366 ($t = 1.07$). None is significant at the 5% level. The profitability loading, -0.350 with $t = -1.88$, comes closest to conventional significance. If a negative loading were stable and statistically reliable, it would mean that Fund X tends to move against the high-minus-low ROE factor rather than earning its returns by harvesting the profitability premium. Here the evidence is suggestive, not conclusive.

The CAPM explains only 0.6% of monthly variation, compared with 4.0% for the q-factor model and 6.9% for q5. Adjusted R^2 values are -0.005, -0.002, and 0.017, respectively. The negative adjusted values in the first two models are valid: after penalising additional parameters, those specifications do not improve on an intercept-only benchmark. These figures are far below the 85–95% market-model R^2 often seen for diversified equity funds. The q factors therefore explain little of Fund X's month-to-month return variation. Low explanatory power is not itself proof of skill, but together with the large intercept it shows that these standard linear equity factors do not account for the reported performance.

The main conclusion is that Fund X's alpha survives both the q-factor and q5 controls. Within this sample and model class, its excess return remains largely unexplained by known systematic equity-factor premia.

Assumption-Aware Diagnostic Assessment

The required $L = 4$ estimates are the primary results. The literal bandwidth expression $4(T/100)^{2/9}$ equals 3.9639 at $T = 96$ and would floor to 3, so $L = 3$ is also checked. The q5 alpha t-statistic is 5.78 at $L = 3$, 5.48 at $L = 4$, 5.07 at $L = 6$, and 4.34 at $L = 12$. The conclusion is not driven by the lag choice.

The q5 residuals remain serially dependent: the Ljung–Box statistic through 12 lags is 51.38 with $p < 0.001$. They are also strongly non-normal (Jarque–Bera $p < 0.001$), and the Breusch–Pagan test is borderline ($p = 0.058$). These findings justify HAC inference and caution against relying on textbook iid OLS standard errors. A 6-month circular block-pairs bootstrap, which preserves local dependence and the joint factor-return observations, gives a 95% q5 alpha interval of [1.85%, 4.10%].

March 2020 is the most influential observation (Cook's distance 1.28). Excluding it leaves a q5 alpha of 2.77% with $t = 5.57$. Across all 96 leave-one-out regressions, alpha ranges only from 2.77% to 3.04%.

Parameter stability is the main qualification. A CUSUM test rejects a constant q5 coefficient vector ($p = 0.0002$). In an objective midpoint split, q5 alpha is 3.92% ($t = 6.01$) in 2017–2020 and 1.74% ($t = 3.31$) in 2021–2024. The alpha remains positive, but its level is not stable. Newey–West inference cannot correct changing coefficients, omitted factors, nonlinear exposures, or a mismatch between a US equity factor model and the fund's underlying strategy. The result should therefore be stated as a large return unexplained by these models, not as proof that every possible risk exposure has been ruled out.

Rolling q5 Alpha: 48-Month Windows

Table 3: Rolling 48-Month q5 Alpha Estimates

Window	N	Monthly α	NW t-stat	R^2
2017-2020	48	3.922%	6.01	0.404
2018-2021	48	4.376%	5.96	0.181
2019-2022	48	3.103%	4.13	0.091
2020-2023	48	2.647%	3.30	0.099
2021-2024	48	1.739%	3.31	0.179

Note: Each regression uses Fund X excess returns and all five q5 factors. Newey–West HAC inference uses a Bartlett kernel with $L = 4$. Adjacent windows share 36 of 48 months, so the estimates are highly correlated and are not five independent tests.

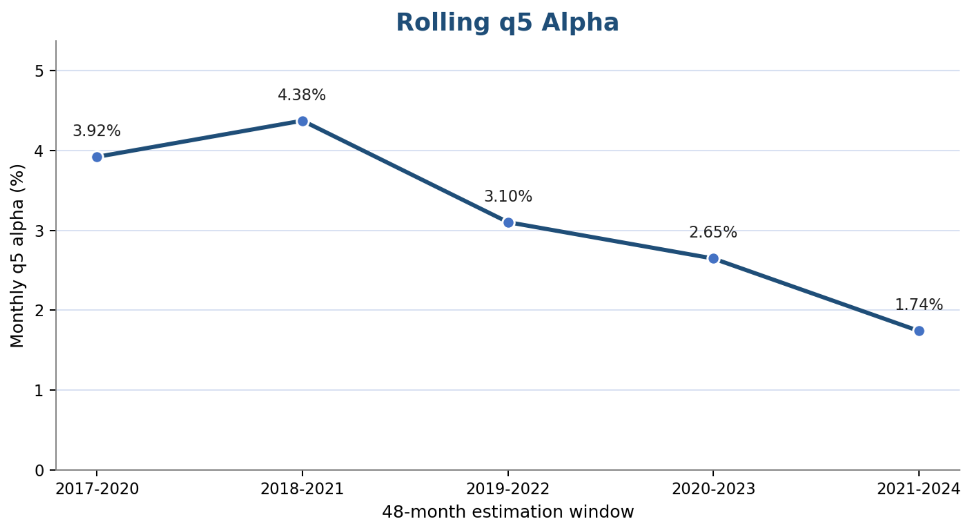

Figure 1: Monthly q5 alpha across overlapping 48-month windows

The rolling sequence is 3.92%, 4.38%, 3.10%, 2.65%, and 1.74%. Alpha first rises in the 2018–2021 window and then declines in each of the next three windows. This does not resemble a single clean shift from a stable 4% level to a stable 2% level. It is more consistent with progressive decay after an early peak. The estimates remain positive and statistically significant in every window. Rolling R^2 ranges from 0.091 to 0.404, showing that the q5 model's ability to explain monthly variation also changes materially over time. Because the windows overlap heavily, this exercise localises the instability descriptively; it is not a formal estimator of a unique break date.

References

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